

Pulkit Tiwari

tpulkit87@gmail.com | linkedin.com/in/pulkittiwari51 | github.com/PulkitTiwari87 | Portfolio

Summary

Computer Science undergraduate specializing in Cybersecurity and Forensics, with experience building Generative AI systems, distributed backend platforms, security automation, and full-stack applications through internships, freelance work, and projects. Strong foundation in data structures, algorithms, machine learning, and LLM applications.

Education

University of Petroleum and Energy Studies (UPES), Dehradun

Aug 2023 – May 2027

B.Tech Computer Science & Engineering – Cybersecurity and Forensics

Skills

Languages & Core CS: Python, Java, C, JavaScript, TypeScript, SQL; Data Structures & Algorithms, Object-Oriented Design, Operating Systems, Computer Networks, Distributed Systems

AI/ML/GenAI: LLMs, RAG, Agentic AI, Prompt Engineering, Embeddings, NLP, XGBoost, Scikit-learn, TensorFlow, CrewAI, AWS Bedrock, AI Guardrails, Qdrant, FAISS

Backend & Data: Node.js, Express.js, REST APIs, gRPC, Microservices, JWT, OAuth, PostgreSQL, MongoDB, MySQL, Redis

Frontend & Cloud/DevOps: React.js, Next.js, Tailwind CSS, Vite; AWS, Docker, Git, GitHub Actions, Vercel, CI/CD

Cybersecurity: Wazuh, TheHive, Cortex, MISP, ELK Stack, Kali Linux, Burp Suite, Nmap, OWASP

Experience

Software Engineering Intern, Ascendion, Bengaluru

Jun 2026 – Sep 2026

- Engineered a 3-agent CrewAI pipeline on AWS Bedrock using Claude models to automate resume reformatting through structured extraction, rendering, and Azure Blob Storage upload with SAS-token authentication.
- Developed an output-validation agent using bias guardrails and knowledge-base-grounded scoring to automate compliance checks across 10+ Generative AI applications and generate standardized branded reports.

Freelance Software Developer, Upwork

Mar 2026 – May 2026

- Delivered end-to-end AI-powered SaaS and automation solutions with React.js, Node.js, Python, PostgreSQL, and LLM APIs, including MomPlan (an LLM-based benefits-eligibility platform) and an AI receptionist prototype for automated lead handling.

Full Stack Developer Intern, Weblicious, Dehradun

Jun 2025 – Jul 2025

- Architected client-facing web applications using React.js, Node.js, and MongoDB with 10+ schemas and an API-first architecture.
- Optimized REST APIs and backend workflows, improving data accessibility and reducing API response time by 25%.

Projects

SOAR Intelligence – AI-Assisted Security Operations Platform

- Built an event-driven ML and microservices security platform on Wazuh, Cortex, TheHive, MISP, ELK, Redis, and Docker for automated detection, alert triage, orchestration, and response.
- Trained an XGBoost alert-triage model on 45,000 records (15,000 synthetic, 30,000 KDD Cup '99) reaching 92.41% accuracy, 0.98 precision, and 0.92 recall; added phishing, anomaly, log-clustering, and blast-radius modules.

Atlas – Fault-Tolerant Distributed Compute Platform

- Architected a Python job-execution platform with a priority scheduler, concurrent workers, and PostgreSQL-backed job state; designed REST/gRPC APIs for scheduling, execution, and monitoring.
- Implemented and benchmarked 4 fault-tolerance mechanisms: heartbeats, failure detection, automatic retries, and graceful recovery.

Real-Time CRAG – Corrective Retrieval-Augmented Generation

- Designed a provider-agnostic Corrective RAG system combining retrieval, relevance evaluation, query refinement, web search, and grounded generation with Qdrant, FAISS, embeddings, and Hugging Face inference.

G-DriveX – Multi-Account Cloud Storage SaaS (live: g-drive-x.vercel.app)

- Built a React/Node.js SaaS unifying multiple Google Drive accounts with server-side Drive-to-Drive streaming and bulk transfers; encrypted OAuth refresh tokens at rest.
- Found and fixed IDOR and OAuth CSRF vulnerabilities; shipped with 24/24 backend tests passing and CI/CD validation.

NEO-Hazard-AI – NASA Near-Earth Object Hazard Classification (live: neo-hazard-ai.vercel.app)

- Leakage-aware ML pipeline on real NASA/JPL CNEOS data (Logistic Regression, Random Forest, XGBoost) with 5-fold CV, holdout evaluation, calibration, SHAP explainability, and anomaly detection.

Positions of Responsibility & Certifications

Public Relations & Sponsorship Head – UPES CSA Student Chapter (Cloud Security Alliance)

Sep 2024 – Apr 2026

- Led a team of 30+ organizing university technical events including hackathons, WebGenesis, and Funtopia 5.0; directed sponsorship outreach for AWS Community Day Dehradun 2025 (400+ attendees), securing industry collaboration including GitHub.

Certifications: Google Cybersecurity (Foundations of Cybersecurity; Play It Safe; Connect and Protect; Tools of the Trade: Linux and SQL); HackerRank – JavaScript (Intermediate), React, Python, CSS.